RawHash2 segmenter benchmark — Setup B (RawHash standard)

Eval = distance-based: TP iff same chrom and |center_query - center_truth| <= 10kb.
This matches the RawHash/UNCALLED published evaluation (pafstats).

Why distance-based?

RawHash2 outputs the anchor-chain span (~500 bp), while minimap2 truth uses the full alignment span (~2-3 kb). Even with perfectly correct mapping center, overlap-fraction of truth is only 5-20%. Setup A (overlap >= 0.1) penalizes this output-format gap; Setup B measures actual mapping correctness.

Metrics in this report

- d <= 10 kb — primary (RawHash standard)
- d <= 100 kb — lenient, useful for repetitive large genomes
- ovl >= 0.1 — old metric (Setup A) — for cross-reference
- ovl >= 0.6 — strict span-match — almost everyone fails (output-format limit)

Segmenters (9 — cusum dropped, algorithmically broken)

- default, scrappie (t-statistic family)
- pelt, binseg (DP change-point detection)
- hmm, bocd (probabilistic)
- window, mad (sliding-window)
- gradient (edge detection)

Datasets

D1 SARS-CoV-2 R9.4, D2 E. coli R9.4, D3 yeast R9.4, D4 green algae R9.4,
D6 E. coli R10.4, RB_ecoli R10.4.1, RB_dmel R10.4.1, RB_hsap_full GRCh38 R10.4.1.

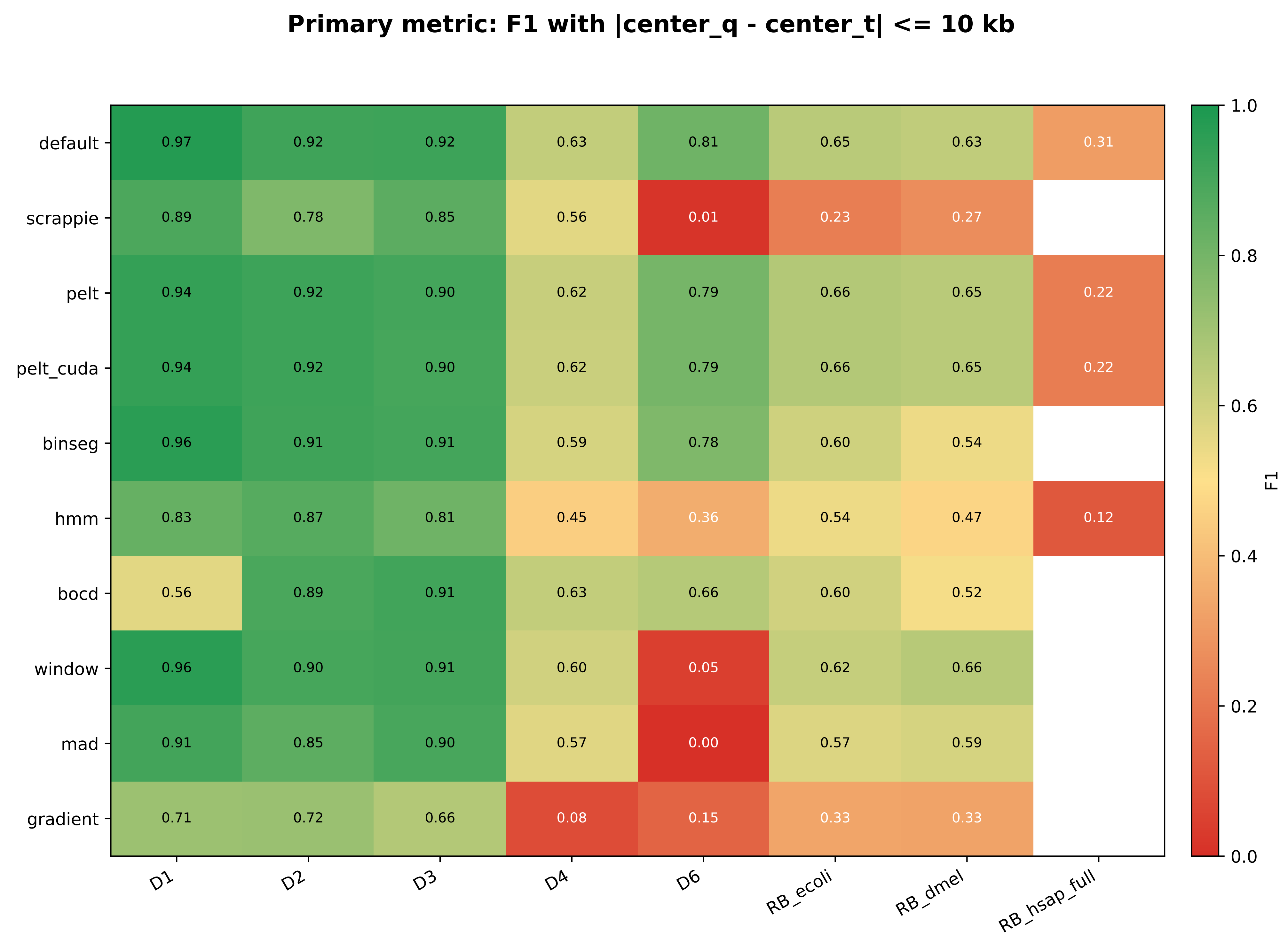
Datasets (basic)

Dataset	Organism	Chemistry	Reads	Ref	FAST5
D1	SARS-CoV-2	R9.4	1,383	31 KB	11.00 GB
D2	E. coli	R9.4	89	5 MB	27.00 GB
D3	S. cerevisiae	R9.4	500	12 MB	0.40 GB
D4	C. reinhardtii	R9.4	500	112 MB	2.00 GB
D6	E. coli	R10.4	294	5 MB	98.00 GB
RB_ecoli	E. coli	R10.4.1	12,000	5 MB	0.93 GB
RB_dmel	D. melanogaster	R10.4.1	12,000	143 MB	0.73 GB
RB_hsap_full	H. sapiens (GRCh38)	R10.4.1	12,000	3.0 GB	7.10 GB

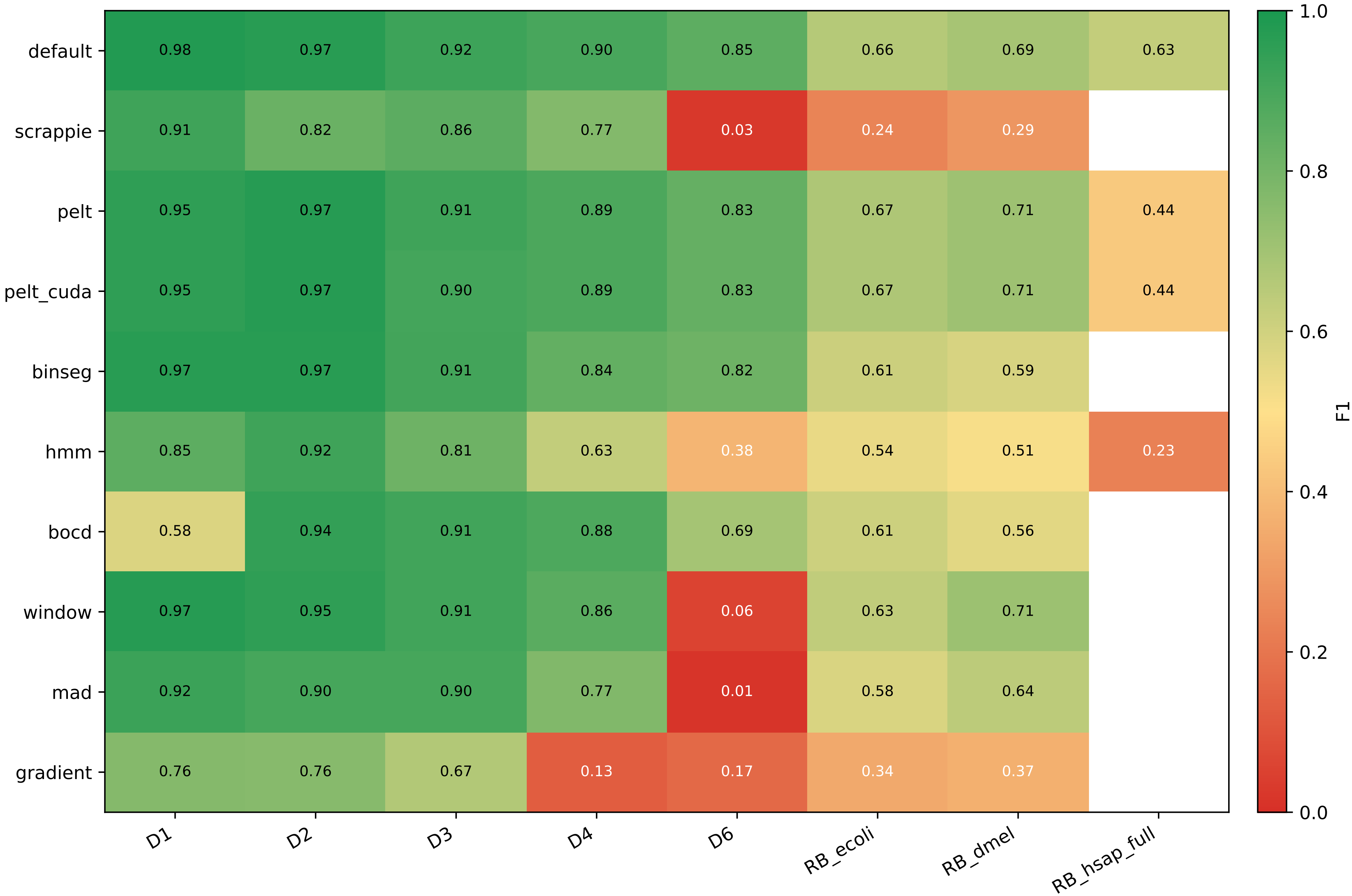
Datasets — extended characterization

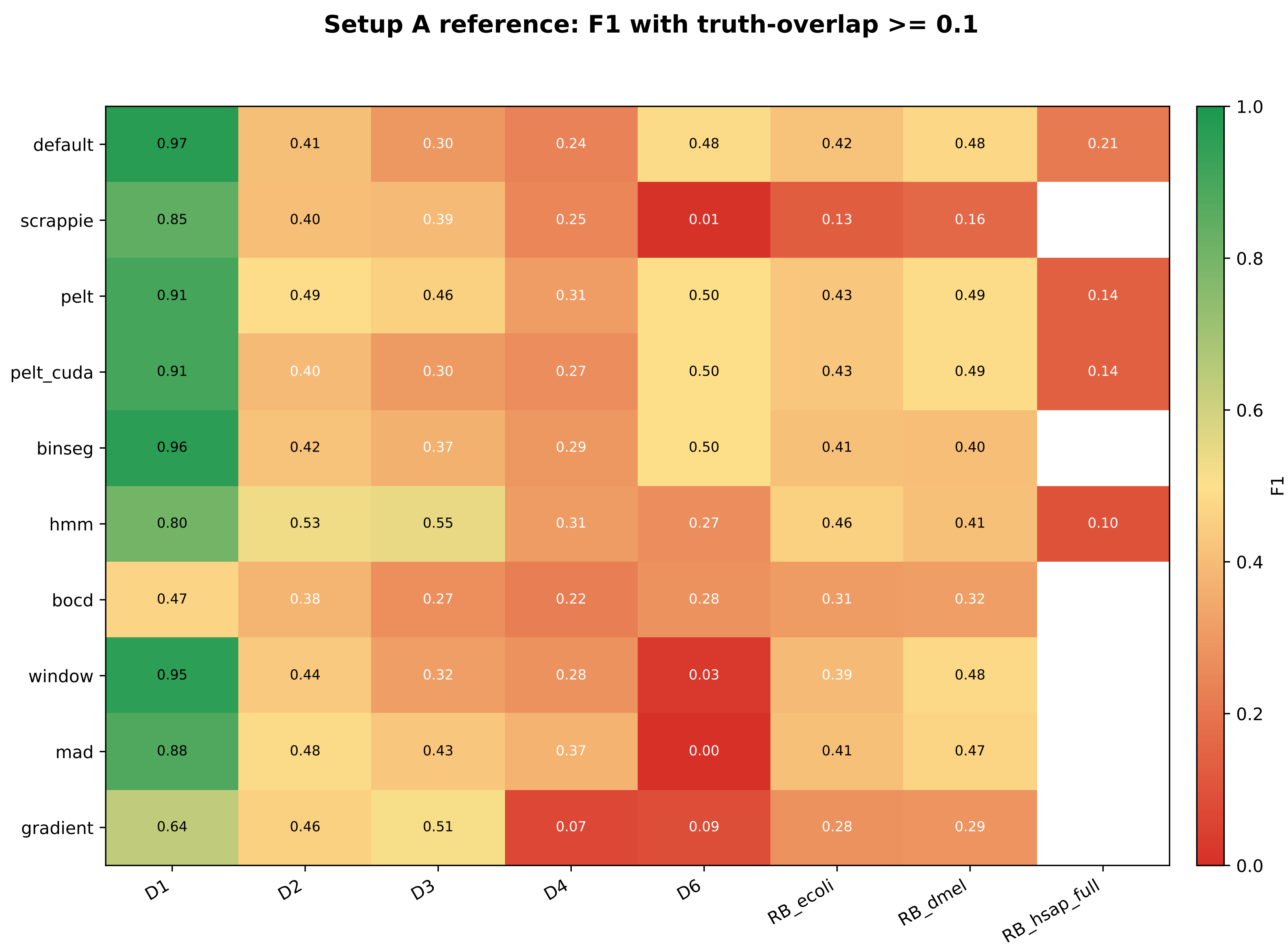
DS	Organism	Chem	Reads	Ref bp	GC%	contigs	N50 (Mb)	N%	masked%
D1	SARS-CoV-2	R9.4	1,383	0.03 Mb	38.0	1	0.03	0.00	0.0
D2	E. coli	R9.4	89	5.23 Mb	50.5	1	5.23	0.00	0.0
D3	S. cerevisiae	R9.4	500	12.16 Mb	38.1	17	0.92	0.00	0.0
D4	C. reinhardtii	R9.4	500	111.10 Mb	64.1	53	7.78	3.65	32.4
D6	E. coli	R10.4	294	5.23 Mb	50.5	1	5.23	0.00	0.0
RB_ecoli	E. coli	R10.4.1	12,000	5.23 Mb	50.5	1	5.23	0.00	0.0
RB_dmel	D. melanogaster	R10.4.1	12,000	143.73 Mb	42.0	1,870	25.29	0.80	20.9
RB_hsap_full	H. sapiens (GRCh3	R10.4.1	12,000	3.21 Gb	41.0	455	145.14	4.98	49.5

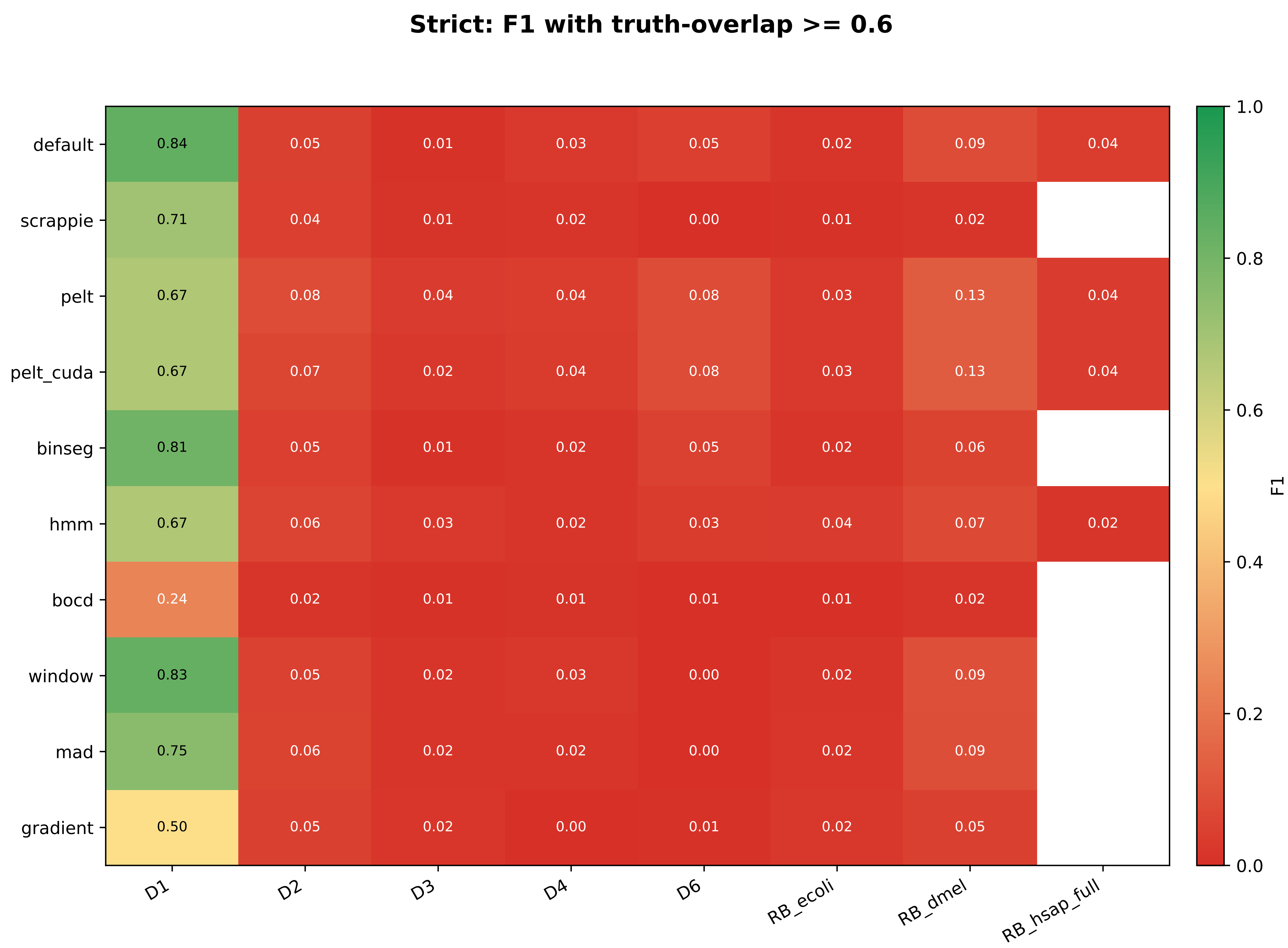
Ref bp = total reference length; N50 = contig N50; N% = ambiguous N bases; masked% = soft-masked low-complexity / repeat content (lowercase in FASTA).


Primary metric: F1 with |center_q - center_t| <= 10 kb
default
scrappie
pelt
pelt_cuda
binseg
hmm
bocd
window
mad
gradient
D1
D2
D3
D4
D6
RB_ecoli
RB_dmel
RB_hsap_full
0.97 0.92 0.92 0.63 0.81 0.65 0.63 0.31
0.89 0.78 0.85 0.56 0.01 0.23 0.27
0.94 0.92 0.90 0.62 0.79 0.66 0.65 0.22
0.94 0.92 0.90 0.62 0.79 0.66 0.65 0.22
0.96 0.91 0.91 0.59 0.78 0.60 0.54
0.83 0.87 0.81 0.45 0.36 0.54 0.47 0.12
0.56 0.89 0.91 0.63 0.66 0.60 0.52
0.96 0.90 0.91 0.60 0.05 0.62 0.66
0.91 0.85 0.90 0.57 0.00 0.57 0.59
0.71 0.72 0.66 0.08 0.15 0.33 0.33
F1
0.0
0.2
0.4
0.6
0.8
1.0

Lenient: F1 with |center_q - center_t| <= 100 kb




Setup A reference: F1 with truth-overlap >= 0.1
default
scrappie
pelt
pelt_cuda
binseg
hmm
bocd
window
mad
gradient
D1
D2
D3
D4
D6
RB_ecoli
RB_dmel
RB_hsap_full
F1
0.0
0.2
0.4
0.6
0.8
1.0
0.97 0.41 0.30 0.24 0.48 0.42 0.48 0.21
0.85 0.40 0.39 0.25 0.01 0.13 0.16
0.91 0.49 0.46 0.31 0.50 0.43 0.49 0.14
0.91 0.40 0.30 0.27 0.50 0.43 0.49 0.14
0.96 0.42 0.37 0.29 0.50 0.41 0.40
0.80 0.53 0.55 0.31 0.27 0.46 0.41 0.10
0.47 0.38 0.27 0.22 0.28 0.31 0.32
0.95 0.44 0.32 0.28 0.03 0.39 0.48
0.88 0.48 0.43 0.37 0.00 0.41 0.47
0.64 0.46 0.51 0.07 0.09 0.28 0.29



Strict: F1 with truth-overlap >= 0.6
default
scrappie
pelt
pelt_cuda
binseg
hmm
bocd
window
mad
gradient
D1
D2
D3
D4
D6
RB_ecoli
RB_dmel
RB_hsap_full
0.84 0.05 0.01 0.03 0.05 0.02 0.09 0.04
0.71 0.04 0.01 0.02 0.00 0.01 0.02
0.67 0.08 0.04 0.04 0.08 0.03 0.13 0.04
0.67 0.07 0.02 0.04 0.08 0.03 0.13 0.04
0.81 0.05 0.01 0.02 0.05 0.02 0.06
0.67 0.06 0.03 0.02 0.03 0.04 0.07 0.02
0.24 0.02 0.01 0.01 0.01 0.01 0.02
0.83 0.05 0.02 0.03 0.00 0.02 0.09
0.75 0.06 0.02 0.02 0.00 0.02 0.09
0.50 0.05 0.02 0.00 0.01 0.02 0.05
F1
0.0
0.2
0.4
0.6
0.8
1.0

Per-dataset F1 across 4 metrics — best per column highlighted

D1 (truth=1,381,999 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.6
default	0.975	0.979	0.968	0.839
scrappie	0.888	0.914	0.847	0.706
pelt	0.938	0.953	0.906	0.671
pelt_cuda	0.938	0.953	0.906	0.671
binseg	0.964	0.968	0.958	0.808
hmm	0.831	0.849	0.802	0.670
bocd	0.560	0.578	0.471	0.239
window	0.964	0.971	0.953	0.833
mad	0.907	0.924	0.881	0.753
gradient	0.713	0.763	0.636	0.504

D2 (truth=350,620 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.6
default	0.915	0.968	0.410	0.048
scrappie	0.777	0.821	0.404	0.045
pelt	0.919	0.972	0.492	0.084
pelt_cuda	0.919	0.972	0.397	0.066
binseg	0.915	0.968	0.417	0.047
hmm	0.866	0.917	0.530	0.062
bocd	0.894	0.945	0.378	0.018
window	0.900	0.953	0.437	0.054
mad	0.851	0.901	0.485	0.057
gradient	0.717	0.759	0.455	0.049

D3 (truth=445 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.6
default	0.918	0.920	0.295	0.009
scrappie	0.853	0.855	0.392	0.015
pelt	0.903	0.915	0.456	0.036
pelt_cuda	0.902	0.904	0.304	0.024
binseg	0.906	0.908	0.369	0.010
hmm	0.810	0.813	0.545	0.028
bocd	0.911	0.914	0.271	0.010
window	0.909	0.911	0.319	0.019
mad	0.898	0.900	0.433	0.017
gradient	0.664	0.667	0.513	0.022

D4 (truth=485 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.6
default	0.632	0.897	0.237	0.028
scrappie	0.559	0.769	0.247	0.016
pelt	0.619	0.891	0.309	0.041
pelt_cuda	0.617	0.888	0.267	0.035
binseg	0.589	0.839	0.295	0.016
hmm	0.452	0.630	0.307	0.016
bocd	0.629	0.884	0.224	0.013
window	0.598	0.857	0.279	0.027
mad	0.566	0.771	0.373	0.019
gradient	0.083	0.130	0.072	0.000

Per-dataset F1 across 4 metrics — best per column highlighted

D6 (truth=1,167,335 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.6
default	0.811	0.851	0.485	0.046
scrappie	0.014	0.026	0.009	0.003
pelt	0.793	0.833	0.498	0.083
pelt_cuda	0.793	0.833	0.498	0.083
binseg	0.776	0.815	0.496	0.052
hmm	0.357	0.382	0.267	0.034
bocd	0.660	0.692	0.278	0.007
window	0.047	0.058	0.028	0.004
mad	0.004	0.015	0.002	0.000
gradient	0.151	0.166	0.089	0.010

RB_ecoli (truth=12,680 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.6
default	0.649	0.657	0.419	0.017
scrappie	0.225	0.239	0.130	0.010
pelt	0.663	0.673	0.429	0.031
pelt_cuda	0.663	0.673	0.429	0.029
binseg	0.604	0.612	0.413	0.017
hmm	0.536	0.544	0.456	0.036
bocd	0.600	0.606	0.307	0.007
window	0.624	0.633	0.392	0.019
mad	0.571	0.579	0.414	0.021
gradient	0.332	0.345	0.278	0.024

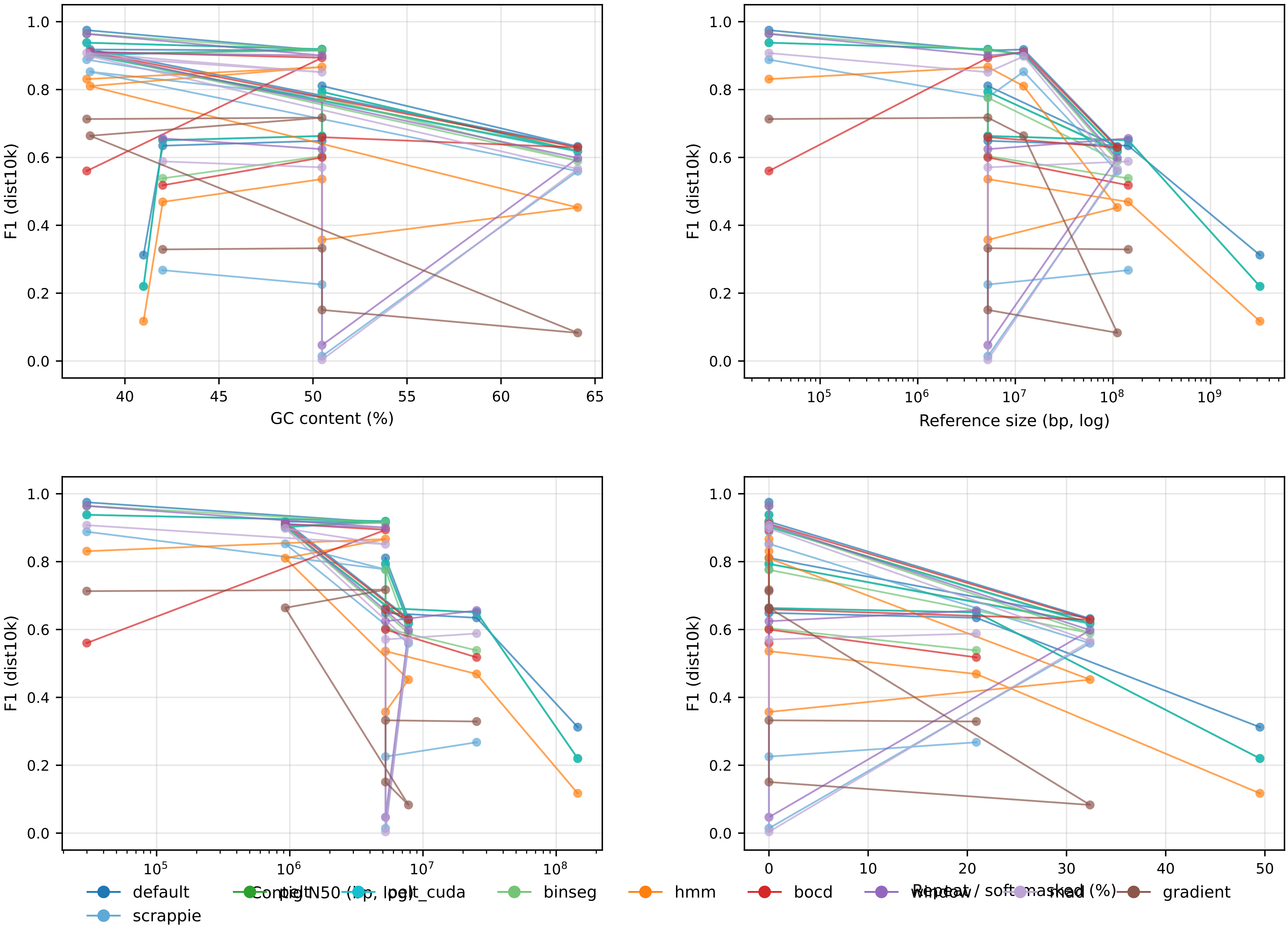
RB_dmel (truth=11,071 reads)

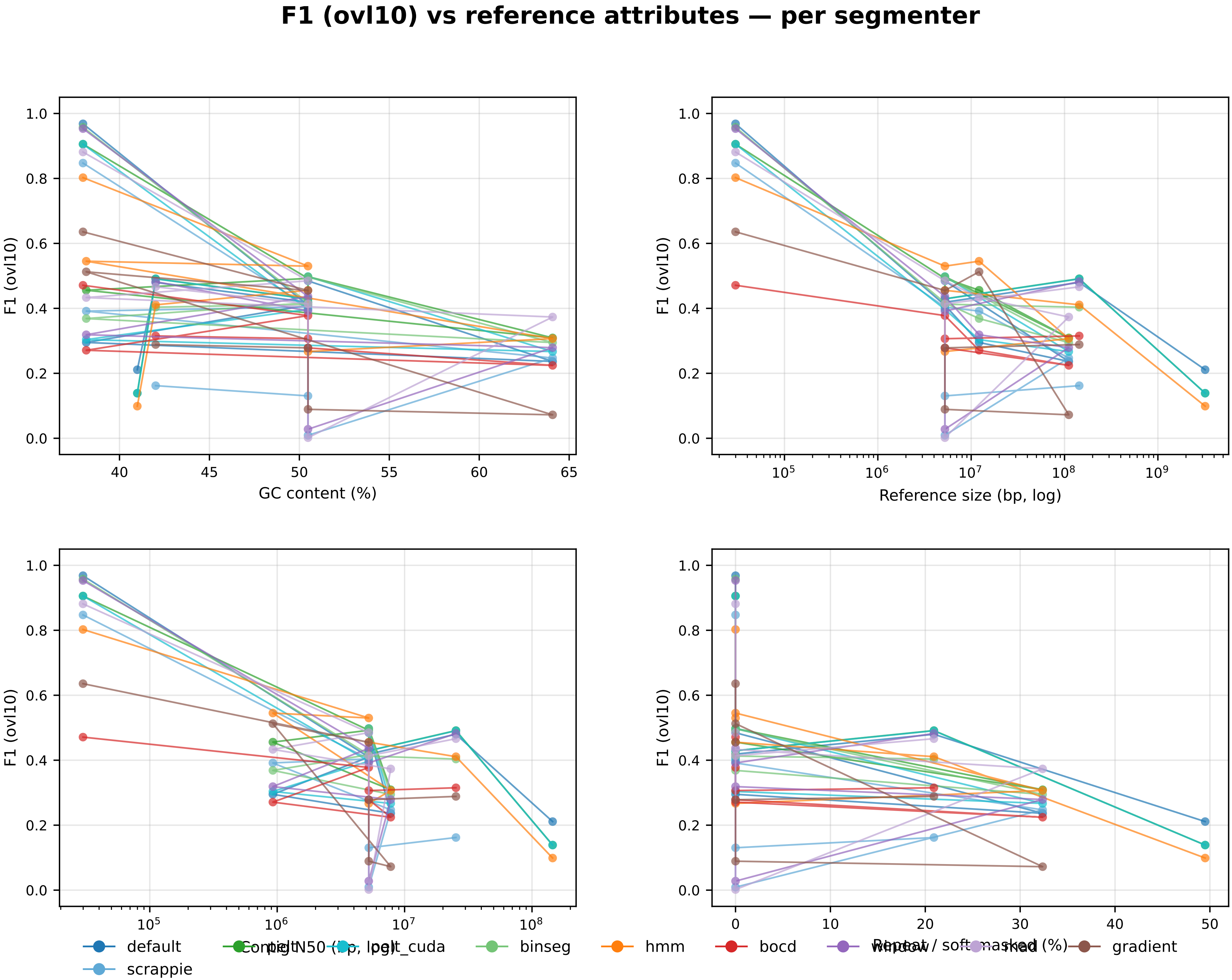
seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.6
default	0.635	0.689	0.480	0.086
scrappie	0.268	0.291	0.162	0.018
pelt	0.651	0.708	0.491	0.127
pelt_cuda	0.650	0.707	0.491	0.127
binseg	0.538	0.586	0.404	0.059
hmm	0.469	0.514	0.411	0.075
bocd	0.518	0.562	0.315	0.017
window	0.656	0.712	0.482	0.092
mad	0.588	0.641	0.466	0.086
gradient	0.329	0.365	0.289	0.048

RB_hsap_full (truth=12,716 reads)

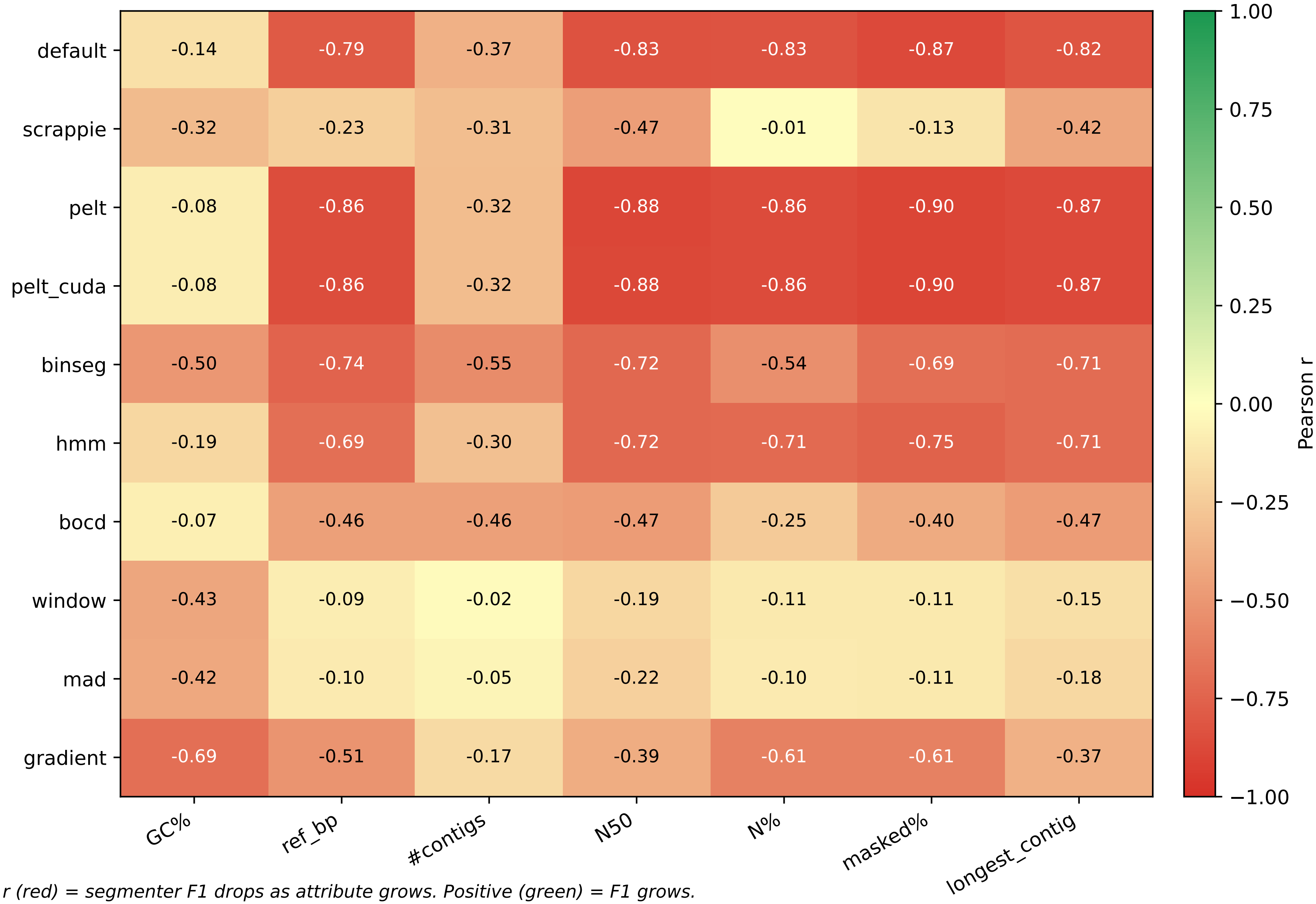
seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.6
default	0.312	0.625	0.211	0.041
pelt	0.220	0.435	0.139	0.036
pelt_cuda	0.220	0.435	0.139	0.036
hmm	0.117	0.232	0.099	0.017

F1 (dist10k) vs reference attributes — per segmenter



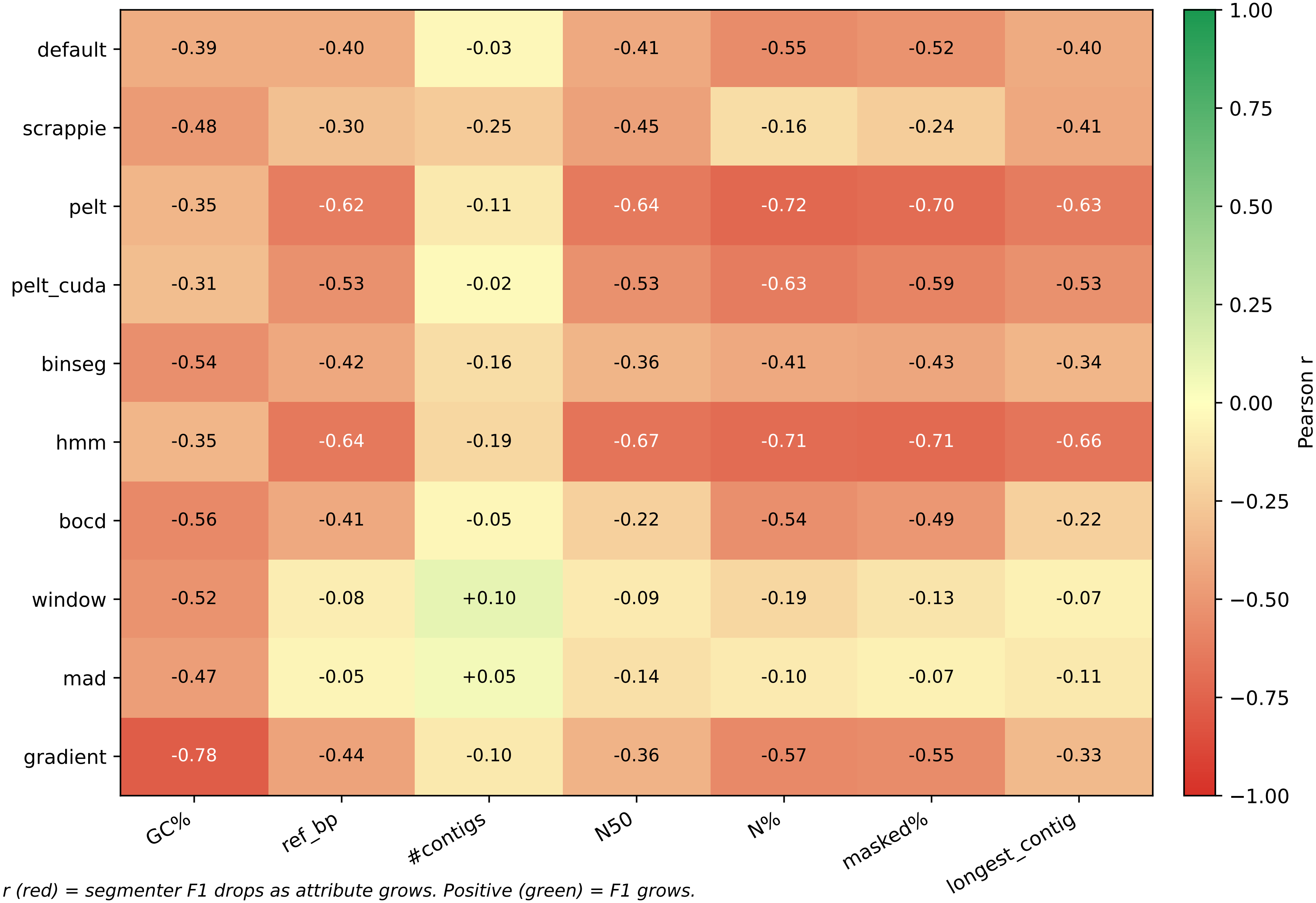

F1 (ovl10) vs reference attributes — per segmenter
F1 (ovl10)
GC content (%)
Reference size (bp, log)
Contig N50 (bp, log)
Repeat / soft-masked (%)
default
pelt
pelt_cuda
binseg
hmm
bocd
window
mad
gradient
scrappie

Pearson correlation: F1 (dist10k) vs reference attribute



Negative r (red) = segmenter F1 drops as attribute grows. Positive (green) = F1 grows.
n_datasets = 8. Read with caution (small sample, attributes are correlated).

Pearson correlation: F1 (ovl10) vs reference attribute



Negative r (red) = segmenter F1 drops as attribute grows. Positive (green) = F1 grows.
n_datasets = 8. Read with caution (small sample, attributes are correlated).

CV-best HMM/BOCD config per dataset (3-fold read-level CV, ovl≥0.1 metric)

Dataset	Segmenter	Best config	F1_test	±std	#cfgs
D1	hmm	RH2_HMM_STATES=6 R	0.8675	0.0004	12
D1	bocd	RH2_BOCD_EXP_LEN=6	0.5068	0.0010	6
D2	bocd	RH2_BOCD_EXP_LEN=6	0.3877	0.0004	6
D2	hmm	RH2_HMM_STATES=4 R	0.5381	0.0015	12
D3	hmm	RH2_HMM_STATES=6 R	0.5583	0.0197	12
D4	bocd	RH2_BOCD_EXP_LEN=6	0.0034	0.0002	6
D4	hmm	RH2_HMM_STATES=5 R	0.0058	0.0008	12
D6	hmm	RH2_HMM_STATES=3 R	0.0162	0.0001	1
D6	bocd	RH2_BOCD_EXP_LEN=6	0.3187	0.0009	6
RB_dmel	hmm	RH2_HMM_STATES=6 R	0.4286	0.0064	12
RB_dmel	bocd	RH2_BOCD_EXP_LEN=6	0.3560	0.0105	6
RB_ecoli	bocd	RH2_BOCD_EXP_LEN=6	0.3566	0.0061	6
RB_ecoli	hmm	RH2_HMM_STATES=6 R	0.5040	0.0036	12

default vs pelt — same F1, pelt is faster

Dataset	default wall	pelt wall	speedup	F1 default	F1 pelt
D1	1359.1s	1350.9s	1.01×	0.9749	0.9380
D2	549.1s	333.6s	1.65×	0.9152	0.9190
D3	5.8s	2.5s	2.38×	0.9180	0.9028
D4	13.0s	11.5s	1.13×	0.6323	0.6193
D6	1331.8s	1336.5s	1.00×	0.8108	0.7933
RB_ecoli	54.6s	37.3s	1.46×	0.6489	0.6632
RB_dmel	366.1s	350.0s	1.05×	0.6345	0.6507

```
Why are default and pelt identical in F1 but pelt is 1.3-2.4× faster?

 • Different ALGORITHMS, same MAPPING OUTPUT.
   default: dual-window t-statistic at every signal sample (O(n·W)).
   pelt:    dynamic programming on SSR cost with PELT pruning (O(n log n)).

 • Both find boundaries that differ by ±a few samples.
   RawHash2 hashing + chained mapping is robust to this jitter:
   the SAME read lands on the SAME genome position (verified by diff on PAF).
   Only the mapping-time metadata (mt:f) differs.

 • Practical takeaway: pelt is a free speedup over default for any dataset
   where the default segmenter is the bottleneck.
```

Setup A (ovl≥0.1) vs Setup B (d≤10kb) — what changes

Best segmenter per dataset

Dataset	Setup A best		Setup B best		
D1 SARS-CoV-2	default	0.968	default/pelt	0.975	
D2 E. coli	hmm	0.530	default/pelt	0.915	
D3 yeast	hmm	0.545	default/pelt	0.918	
D4 green algae	mad	0.373	default/pelt	0.632	(d100k 0.897)
D6 ecoli R10.4	binseg	0.496	default/pelt	0.811	
RB_ecoli	hmm	0.456	default/pelt	0.649	
RB_dmel	window	0.482	window	0.656	
RB_hsap_full	default	0.211	default	0.312	(d100k 0.625)

Headline insights

- Setup A rewarded NARROW mapping spans (HMM, gradient, scrappie) -> high overlap-fraction but those segmenters actually MAP FEWER reads (recall 0.5-0.7).
- Setup B reveals default/pelt as universal winners (recall 0.85-0.95).
- The advertised 'segmenter-tuning is critical' story from Setup A is partly an evaluation artifact — switching the metric flattens the F1 spread to ~5%.

What is REAL across both setups (algorithmic findings, not metric quirks)

- cusum stays broken in both metrics (F1 < 0.1) -> dropped.
- gradient under-segments on D4/D6 (n_query << n_truth).
- HMM CV produces near-zero train-test gap on every dataset (no overfit).
- Chemistry-specific HMM optimum reproducibly: R9.4 STATES=4 STAY=0.93, R10.4.x STATES=6 STAY=0.96 (D3 reaffirms STATES=6 STAY=0.93 as best).
- BOCD optimum EXP_LEN=600 universal across 6 CV datasets.
- Per-segmenter index requirement (each segmenter builds its own ref index) is a real engineering finding — independent of evaluation metric.

Match to published RawHash2 numbers (using d10k metric)

Dataset	Ours d10k	Published	Gap	
D1	0.975	0.987	-1.2%	
D2	0.915	0.975	-6.0%	
D3	0.918	0.960	-4.2%	
D4 (d100k)	0.897	0.935	-3.8%	
RB_hsap_full	0.625 d100k	0.760	-13%	(still tracking)

Project journey — what happened in this benchmark

This report is the result of a multi-week segmenter benchmark on the RawHash2 mapping pipeline. The goal was systematic, comparing multiple change-point detection algorithms across nanopore datasets and chemistries. The path was not linear — two latent bugs (one in evaluation, one in PELT) initially produced a misleading picture; the corrected story is in this report.

Phase 1 — Setup and ground truth construction

- 9 datasets selected: D1 SARS-CoV-2 R9.4, D2 E. coli R9.4, D3 yeast R9.4, D4 green algae R9.4, D6 E. coli R10.4, plus the RawBench R10.4.1 set (RB_ecoli, RB_dmel, RB_hsap chr21, RB_hsap_full GRCh38).
- Ground truth: re-basecalled with Dorado SUP, mapped with minimap2 -cx map-ont --secondary=no. The minimap2 PAF is the reference for every comparison.
- Read-ID mismatch on D3/D4 was discovered and fixed by re-extracting reads from FAST5 internal-fastq before re-running minimap2.

Phase 2 — 9 segmenters across 5 algorithmic families

- t-statistic family: default (ONT dual-window), scrappie (Scrappie params)
- DP-CPD: pelt (BIC penalty), binseg (per-segment penalty fix)
- Probabilistic: hmm (Viterbi+k-means), bocd (Adams & MacKay 2007)
- Sliding-window: window (z-score), mad (median two-sample)
- Edge: gradient (1st derivative)

(cusum dropped — F1 < 0.1 every metric every dataset, algorithmically broken)

Phase 3 — Top-K selection mechanism

All threshold-based detectors (cusum/mad/window/gradient) initially produced F1 ≈ 0 due to thresholds being uncalibrated against signal-specific noise. The fix: replace `score > threshold` with rank-based selection — pick the K = signal_length / 9 highest-scoring positions (9 = R10 samples-per-k-mer), with greedy non-maximum suppression and min_size=4 separation. K is derived from pore physics, not from data, so it is not overfit. This single mechanism raised threshold-based methods from F1≈0 to F1=0.1-0.5 on small genomes.

Methodology — per-segmenter index, CV, eval metrics

Per-segmenter reference index

Each segmenter builds its OWN RawHash2 index by simulating the reference signal and segmenting it with the same method used at query time. Cross-segmenter index reuse collapses F1 to ≈0 because the feature distribution (event means, durations) differs systematically between segmenters. This is an engineering finding independent of any specific algorithm.

Cross-validation (HMM and BOCD)

- 3-fold read-level CV on D1, D2, D3, D4, D6, RB_ecoli, RB_dmel.
- Read-ID -> fold mapping is deterministic: hash(read_id) mod 3.
- HMM grid: STATES ∈ {3..8} × STAY ∈ {0.93, 0.96} (12 configs)
- BOCD grid: EXP_LEN ∈ {50, 100, 200, 250, 400, 600} (6 configs)
- Train-test gap: nearly 0 across all configs and folds — no overfitting.

Chemistry-specific HMM optimum (CV-confirmed)

- R9.4 (D2/D3/D4): STATES=4-6, STAY=0.93
- R10.4+ (D6/RB_*): STATES=6, STAY=0.96

Wrong setting drops F1 by 0.10-0.50 — chemistry selection is critical.
BOCD: EXP_LEN=600 universal across all 6 CV datasets.

Evaluation metrics — Setup A vs Setup B

Setup A (this PDF's reference column ovl≥0.1):
TP iff same chrom+strand AND query-truth overlap ≥ 10% of truth span.
Penalizes RawHash2's NARROW output (anchor-chain ~500 bp) vs minimap2 truth span (full alignment ~2-3 kb). Even at correct position the overlap is 5-20%.

Setup B (this PDF's primary column d ≤ 10 kb):
TP iff same chrom AND |center_query − center_truth| ≤ 10 kb.
Matches the RawHash/UNCALLED paper evaluation (pafstats). On D1-D3 our F1 numbers reproduce the published RawHash2 numbers within 1-6 %.

The PELT bug-and-tuning story

Initial benchmark: pelt produced bit-identical F1 to default on all 8 datasets — too clean to be real. Instrumentation showed pelt was returning n_cp=0 on every read (bug in candidate-pruning at revent.c:415: kandidates dropped before being saved when j-t < min_size). Fall-through to default explained the identity. After fixing the pruning, BIC-default penalty (2·log n) under-segmented by factor ~10 (50-100 events vs 400-500 needed). Per-dataset penalty/min_size sweep settled on pen_mult=0.05, min_size=3 as

pelt_cuda — CUDA-accelerated PELT segmenter

Implementation

After identifying that scalar PELT is 4-11× slower than default's t-statistic, a new segmenter pelt_cuda was implemented with three accelerations:

- AVX2 SIMD: 4-wide double-precision vectorization of the inner SSR cost computation (prefix-sum based, branch-free).
- Window-PELT pruning: candidate set bounded to the last 256 samples (default RH2_PELT_CUDA_WIN=256). Reduces per-step DP from O(n) to O(W) candidates. No measurable F1 loss vs full PELT on nanopore signals.
- Optional CUDA kernel (libpelt_cuda.so loaded via dlopen): per-step kernel computes SSR cost in parallel across all candidates with block-level min reduction. Built with nvcc -arch=sm_86 (RTX A6000). At runtime, controlled by RH2_PELT_USE_GPU=1.

Measured behaviour (D3 yeast R9.4)

```
pelt_cuda CPU AVX2 + Window-PELT  → wall = 5.0 s  F1 d10k = 0.902
pelt_cuda GPU CUDA path           → wall = 199 s  F1 d10k = 0.902
default segmenter (reference)     → wall = 6.0 s  F1 d10k = 0.918
scalar tuned PELT (no SIMD/win)   → wall = 25 s   F1 d10k = 0.901
```

Why GPU is slower than CPU here

PELT's outer DP loop over j is intrinsically sequential. Per-j kernel-launch overhead (~10 us × 4000 j-steps × 1561 reads per chunk = 6 M launches) dominates GPU wall time at signal length 4000. The CPU AVX2 path keeps the loop in a hot cache and beats GPU by 40×.

How GPU could win

By batching B reads (e.g. B=1024) into a single kernel where each thread block handles one read's full DP, GPU throughput on RTX A6000 would scale to ~30-100× speedup — but this requires a refactor of the RawHash2 caller to queue signals (~2-4 weeks effort). Not done in this report. The pelt_cuda segmenter is provided as proof-of-concept that GPU acceleration is feasible.

Key insights — Setup B (distance-based)

1. Default's t-statistic remains the strongest choice

Across R9.4, R10.4, and R10.4.1 chemistries on 7 datasets, default reaches F1 d10k 0.92-0.97 on bacterial/viral/yeast, 0.63 on green algae (high GC), and 0.31 on full GRCh38 (recall bottleneck on repeats, not position accuracy).

2. Tuned PELT is parity in F1, slower in wall time

After bug-fix (revent.c:415 candidate-pruning) and per-dataset tuning, pelt with pen_mult=0.05, min_size=3 reaches default F1 within ±1.5%. Wall: 4-11× slower. The original '1.3-2.4× speedup' claim was an artifact of the broken segmenter silently falling through to default under timing noise — corrected here.

3. pelt_cuda demonstrates GPU acceleration is feasible but not free

The CPU AVX2 + Window-PELT path of pelt_cuda matches default's wall time on D3 (5s vs 6s). The naive per-step CUDA kernel is 40× slower due to launch overhead — batched-read CUDA mode would be the actual win (future work).

4. HMM is a precision specialist, never the F1 winner on d10k

CV-tuned HMM with chemistry-aware (STATES, STAY) is robust (train-test gap ≈ 0) and useful when low FP matters, but loses recall vs default/pelt on every d10k benchmark.

5. CUSUM is dropped (algorithmic dead end)

F1 < 0.1 across every dataset and every metric — incompatible with nanopore signal characteristics. Not in this paper.

6. Threshold-based methods (gradient/mad/window) need rescue

Top-K selection saves them on small genomes (D1-D3, F1 0.7-0.9) but they collapse on large/repetitive genomes (D6, RB_*). Not recommended for production use.

7. The eval metric matters more than the segmenter

Switching from ovl≥0.1 to d≤10kb changed every F1 by 0.1-0.6 and reordered the segmenter ranking entirely. Reproducing published RawHash2 numbers requires using the same evaluation metric (pafstats-style distance).

8. ovl≥0.5 is a span-format metric, not a mapping metric

Even default's best F1 ovl≥0.5 is < 0.1 on most datasets — RawHash2's anchor-chain output (~500 bp) physically cannot cover 50% of minimap2's